

IDC 410: Machine Learning

Spring Semester 2025-26

Assignment 01

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Submission: Before 10th January 2026

1. Take an image of something you like and convert it to a **greyscale** image.
 - Extract the **intensity array** of the greyscale image in a two-dimensional (matrix) and a one-dimensional array
 - Obtain the frequency distribution of the pixel intensity and store it in a one-dimensional array. Plot the histogram of the greyscale pixel intensity values (X-axis: Intensity and Y-axis: Frequency). Compare frequency distribution obtained from histogram with one dimensional frequency distribution of the pixel intensity.

This histogram will show how the different intensity levels are distributed in the image. Additionally, verify that the sum of all histogram frequencies is same as the total number of pixels in the image.

2. Take any grey scale image which can be represented it as a $n \times n$ pixels matrix given below as follows, with each value representing intensity of the pixel.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n_{\text{pixel}}} \\ a_{21} & a_{22} & \cdots & a_{2n_{\text{pixel}}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n_{\text{pixel}}1} & a_{n_{\text{pixel}}2} & \cdots & a_{n_{\text{pixel}}n_{\text{pixel}}} \end{bmatrix}$$

Now select a region ($n \times n$) matrix and then modify it, by say changing its intensity and save the modified matrix as image and then display it to see the changes. This will help you to understand the Intensity array and also how to manipulate the image.

3. Convert the following image into a grey scale image. Carryout reflection operation of pixel intensities along the vertical axis a) on the LHS edge of the image b) on the RHS of the image and c) at the center of the image. Save the image obtained by each of the reflection operation and display it on the screen.



4. Take a clean A4 size paper without any wrinkles on it. Draw a proper circle of radius 50 mm with a pencil and properly darken the inner area of the circle. Take two photographs of the shaded circle using your mobile camera say from a distance of exactly 50 cm and 80 cm. Translate these two images to greyscale images and then obtain the greyscale intensity histograms of both the images. Obtain as many inferences as possible using these images.
5. Explore the possible way to relate optical image and physical parameters of objects in the image.
6. Propose the method to get focal length of mobile camera.